

Symmetric Polynomial Representations of Boolean Functions

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Define 721.2: PolyApprox as the class of polynomial functions that approximate a particular boolean function. They are of the form $p : \mathbf{R} \rightarrow \mathbf{R}$ and have the invariant

$$\forall x \{0, 1\}^n : f(x) = p(x)$$

Define 721.1:

$$V_k = \sum \Pi x_k$$

$$V_k = \sum_{S \subseteq [n]} M_s x_k$$

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Lemma 721.1: Symmetric polynomials have a special sum decomposition

Let p^s be a symmetric function, then we can write

$$isSymmetric(p^s) \Rightarrow p^s(|x\rangle) = \sum V_1(|x\rangle) = a_0 + a_1 V_1 + \dots a_n V_n$$

$$p = \sum_k^{2^n} a_k m_s$$

Lemma: 721.1: Polynomials representing boolean functions have sum decomposition

Any polynomial $p \in \text{PolyApprox}$, can be written as a linear combination of 2^n monomials where each monomial is determined by a subset sampled from $[n]$. Concretely,

$$\forall p \in \text{PolyApprox}: \quad p = \sum_{S \in [n]}^{2^n} \alpha_S \cdot X(m_S)$$

Where $m_s = \sum_{i \in S} f_i$ and $\alpha_s \in \mathbf{R}$ (the scalarcoefficient).

And $f_i: [n] \mapsto \{0, 1\}$ is the function that sends i to 1 and all other values to zero.

Prop: The effect Permuting a PolyApprox has on it's sum decomposition

Let $p \in \text{PolyApprox}$, then by lemma 721.1, it has a decomposition $p() = \sum_{S \in [n]}^{2^n} \alpha_S \cdot X(m_S)$.

Let $\pi \in S_n$

$$p(\pi x) = \sum_{S \in [n]}^{2^n} \alpha_S \cdot X(m_S)$$

Here we consider $x \in [M]^n$ as functions $[n] \rightarrow [m]$ Example:

$n = 2 \quad m = 2$

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Theorem 721.1: Any Symmetric ApproxPoly can be made univariate at no cost

Let S be the symmetrization function. That is, S takes any multivariate polynomial $p \in R[X_1 \dots X_n]$ into the symmetric polynomial p^{symm} .

Let $H: \{0, 1\}^n \rightarrow n$ be a function that computes the hamming weight of an n bit vector. Then we claim

$$p \in R[X_1, \dots, X_n] \implies \exists q \in R[X], \forall \vec{\alpha} = (\alpha_1, \alpha_n) \in \{0, 1\}^n \quad q(H(\vec{\alpha})) = p^{symm}(\vec{\alpha})$$

Proof:

Let p^{symm} be the symmetrization of p . That is we can write $S(p) = p^{symm}$.

So p^s is symmetric and therefore satisfies

$$\forall \pi_k \in S_n: \quad p(\pi_k |x\rangle) = p(|x\rangle)$$

Lemma 1: